

Life in mountains

Life at high altitude can be harsh. Food is often scarce, the weather can be treacherous, and the thin air can make it difficult to breathe. However, there is also more space, relatively little interference from humans, and fewer predators than there are lower down. Many animals are “incomers,” using high ground as an extension of their normal range, but there are also some that live only on high ground. In large mountain chains, many animals have a wide distribution; but isolated peaks are often inhabited by animals that are found nowhere else.

Breathing thin air

At 19,800 ft (6,000 m), air is half as dense as it is at sea level. As a result, it contains only half the normal amount of oxygen—so little that anyone trying to breathe at this height would have difficulty remaining conscious. Yet some mountain animals live even higher than this because they have evolved specialized body systems that enable them to get the maximum amount of oxygen into their blood.

In the vertebrate world, birds are the unrivaled experts at high-altitude living. This is because air passes through their lungs in only one direction, not in and out, which ensures that a high proportion of the air's oxygen enters the blood—far more than enters a mammal's bloodstream in the same conditions. This fact is apparent from the height at which birds are capable of flying. In the Himalayas, choughs have been seen fluttering around campsites at over 26,400 ft (8,000 m), and there are records of vultures colliding with planes at over 36,300 ft (11,000 m)—far higher than Mount Everest.

Birds are unusual in being able to cope with rapid changes in altitude without experiencing any ill effects. For mammals, moving from one elevation to another necessitates special adjustment by the body, which is achieved by acclimatization—a process that can take several weeks to complete. During acclimatization, the number of red cells in the blood slowly increases, boosting its oxygen-carrying capacity. This physical adjustment, which is shown by a broad range of mammals—including humans—is temporary. If an animal moves back to lower ground, the process is reversed.

However, for mountain mammals, such as the vicuna and ibex, adaptation to life high up is a permanent state, not something that can be switched on and off. When measured as a proportion of volume, vicunas have 3 times more red cells in their blood than most other mammals, and the hemoglobin in their red cells is unusually good at collecting oxygen. As a result, vicunas can run almost effortlessly on the altiplano—the high-elevation plateau that runs the length of the Andes.

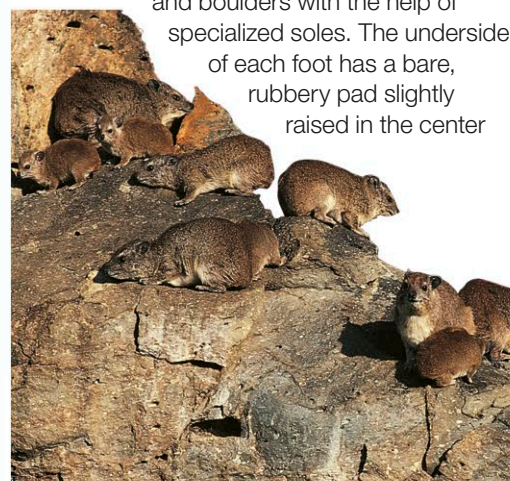
Compared with mammals, cold-blooded animals, such as reptiles, have fewer problems with thin air because they use oxygen more slowly. For them, the main problem with mountain life is cold: if the temperature is too low, their body processes slow down, and their muscles have difficulty working.

Movement

Mountains seem almost purposely made for soaring birds because strong air currents make it easy for them to gain height. For animals on or close to the ground, moving around is not so easy. Many insects are wingless, and species that do fly usually keep close to the rock to reduce the risk of being blown away by the wind.

For larger animals, the situation is even more hazardous, for a single misjudged move can lead to a fatal fall. Many rock-dwelling mammals therefore have feet designed to prevent slipping. In Africa and the Middle East, hyraxes—rabbit-sized grazing mammals—run over rocks

and boulders with the help of specialized soles. The underside of each foot has a bare, rubbery pad slightly raised in the center



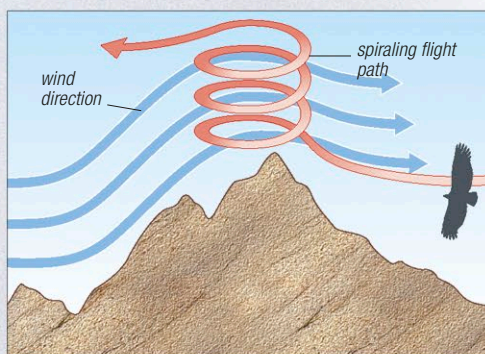
NONSLIP FEET

The hyrax's small, adhesive feet are perfect for negotiating rock, although some species also use them for climbing trees. In addition to living high up, rock hyraxes inhabit kopjes—miniature mountains of eroded boulders scattered on the African savanna.

and moistened by sweat. As a result, each foot works like a suction cup so the hyrax can cling to the surface of smooth rocks without slipping.

Hooves may seem far from ideal for climbing, and it is true that some hoofed mammals, such as horses, have great difficulty moving around on rocky slopes. But in climbing mammals, such as mountain goats and klipspringers, hooves have evolved into perfect aids for moving about in mountains: they are small and compact, allowing them to fit onto narrow ledges, and they have hard edges surrounding rough, nonslip pads. These combined characteristics make for good grip in all conditions, including rain and snow.

Getting a firm grip is essential for moving on rock, but equally important is a strong sense of balance and head for heights. Most terrestrial mammals are instinctively afraid of steep drops but, from an early age, mountain dwellers show what appears to be a reckless disregard for their own safety. Adult chamois take 20 ft (6 m) leaps, and can run down nearly vertical slopes



SOARING

Updrafts provide an almost effortless means of travel for birds like the Andean condor, which can cover hundreds of miles in a day by riding waves of rising air.